

12.421 / 12.621 Lecture 12

Atmospheric scattering and black body radiation
(or Why is the sky blue?)

Topics covered

- Rayleigh scattering
- Mie scattering
- Planck's Law

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Rayleigh Scattering

- Describes scattering from spherical particles whose radius $r \ll \lambda$ ($\lambda \equiv$ EM wavelength)

$\Rightarrow$ Phase of EM wave $\approx$ constant

- This is simplest and very important models of EM scatter $\rightarrow$ major implications for RS

- Derivation

- Consider small, homogeneous, isotropic, spherical particle w/ $r \ll \lambda$ (assume molecule-size particle)

- Define incident E field $\underline{E}_0$ and let $\underline{E}$ be total field ($\underline{E} = \underline{E}_0 + \text{E field caused by electric dipole}$)

- Define induced dipole moment

$$\underline{P}_0 = \alpha \underline{E}_0$$

where

$\alpha \equiv$ polarizability - an intrinsic material property that is related to the dielectric constant

$\hookrightarrow$ In general, α is a 2nd-order tensor but can often be assumed to be a scalar
 $\rightarrow$ we will take the scalar approximation

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- As we discussed way back in Lecture 1, applied electric field causes dipole to oscillate in a plane that is normal to propagation vector because EM waves are transverse waves ($\underline{E}$, $\underline{H}$, and $\hat{k}$ are mutually orthogonal)
- Oscillating dipole emits EM radiation $\rightarrow$ "the scattered wave"
- It can be shown (see Griffiths 2013) that dipole field at distance R from dipole can be represented as sum of
 - 1) Static field $\underline{E}_S \propto \frac{1}{R^3} \underline{p}$
 - 2) Induction field $\underline{E}_I \propto \frac{1}{R^2} \frac{d\underline{p}}{dt}$
 - 3) Radiation field $\underline{E}_R \propto \frac{1}{R} \frac{d^2 \underline{p}}{dt^2}$
- In far-field ($R \gg \lambda$), radiation dominates, and it can be shown

$$\underline{E} = \frac{1}{c^2 R} \frac{\partial^2 \underline{p}}{\partial t^2} \sin \theta$$

where $\theta \equiv$ angle between scattered dipole moment $\underline{p}$ and direction of observation

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- Scattered dipole moment can be written

$$\underline{p} = \underline{p}_0 e^{j(\omega t - KR)}$$

- Thus, total E-field is

$$\underline{E} = -E_0 \frac{K^2 \alpha}{R} \sin \theta e^{j(\omega t - KR)}$$

- Recall that we can represent any $\underline{E}$ field by orthogonal EM wave polarizations h and v (a.k.a., perpendicular and parallel), thus

$$E_h = -E_0 \frac{K^2 \alpha}{R} e^{j(\omega t - KR)} \sin \theta_1$$

$$E_v = -E_0 \frac{K^2 \alpha}{R} e^{j(\omega t - KR)} \sin \theta_2$$

where $\theta_1 = \frac{\pi}{2}$ by definition (see lecture notes on geometric optics)

and $\theta_2 = \frac{\pi}{2} - \phi$ where $\phi \equiv$ scattering angle (angle between incident & scattered waves)

- We are interested in the intensity I , defined as power per unit area, which is related to E-field magnitude as

$$I_p \propto |E_0|^2$$

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- Therefore

$$I_h = I_{0h} \frac{K^4 \alpha^2}{R^2}$$

$$I_v = I_{0v} \frac{K^4 \alpha^2}{R^2} \cos^2 \phi$$

$$\Rightarrow I = I_h + I_v = (I_{0h} + I_{0v} \cos^2 \phi) \frac{K^4 \alpha^2}{R^2}$$

- Let us consider case of scattering from Earth's atmosphere $\rightarrow$ sunlight is unpolarized $\Rightarrow I_{0h} = I_{0v} = \frac{I_0}{2}$

where $I_0 \equiv$ incident intensity

$$\Rightarrow \boxed{\begin{aligned} I &= I_0 \frac{\alpha^2}{R^2} K^4 \frac{1 + \cos^2 \phi}{2} \\ &= I_0 \frac{\alpha^2}{R^2} \left(\frac{\omega}{c} \right)^4 \frac{1 + \cos^2 \phi}{2} \end{aligned}}$$

- This is the formula originally derived by Rayleigh in 1871

- Take-homes

• Scattered intensity $\propto R^{-2}$ (due to spreading losses)

• Scattered intensity $\propto \omega^4$ (equiv. $\frac{1}{\lambda^4}$)

$\Rightarrow$ higher frequency radiation scatters more strongly

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- Scattered intensity $\propto \alpha^2 \Rightarrow$ related to dielectric constant ϵ of atmosphere; can be shown that:

$$\alpha \propto \left(\frac{\epsilon - 1}{\epsilon + 2} \right) \quad \left(\text{"Lorentz-Lorentz formula"} \right)$$

- $\Rightarrow$ Intensity of scattered light increases $\propto 1/\epsilon$
- $\Rightarrow$ moisture influences scattered intensity even in clear skies (other factors such as particulates will play an important role, too, as we will see in a bit)

- Polarization of light: dependence on scattering angle ϕ only comes into vertically polarized term; $I_v \propto \cos^2 \phi \Rightarrow \phi = \frac{\pi}{2}$ has only horizontally polarized light
- $\Rightarrow$ atmosphere acts as a polarizing filter, something that photographers who use polarizing filters will know very well

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Rayleigh scattering

- An extension of the ∇ model we just derived that accounts for particle size r ($r \ll \lambda$) gives

$$I = I_0 \left(\frac{\omega}{c} \right)^4 \frac{r^6}{R^2} \frac{1 + \cos^2 \phi}{2} \left(\frac{n^2 - 1}{n^2 + 2} \right)^2$$

where $n^2 = \epsilon$ ($n = \text{index of refraction}$)

$\Rightarrow$ Scattered intensity scales as $I \propto r^6$ (for $r \ll \lambda$)

- What about when $r \gtrsim \lambda$?

Mie - scattering

- Key approximation in Rayleigh scattering is that $r \ll \lambda$ meaning we can ignore the E field inside the particle and approximate scattered radiation as having same phase $\rightarrow$ works for clear atmosphere
- As $r \rightarrow \lambda$, we have to allow EM wave phase to vary over surface and for particle to contain E -fields $\Rightarrow$ interference
 $\rightarrow$ generally the case when particles or aerosols are present
- Formal derivation of Mie scattering beyond the scope of this course

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- It can be shown that the scattered intensity for any size spherical particle is

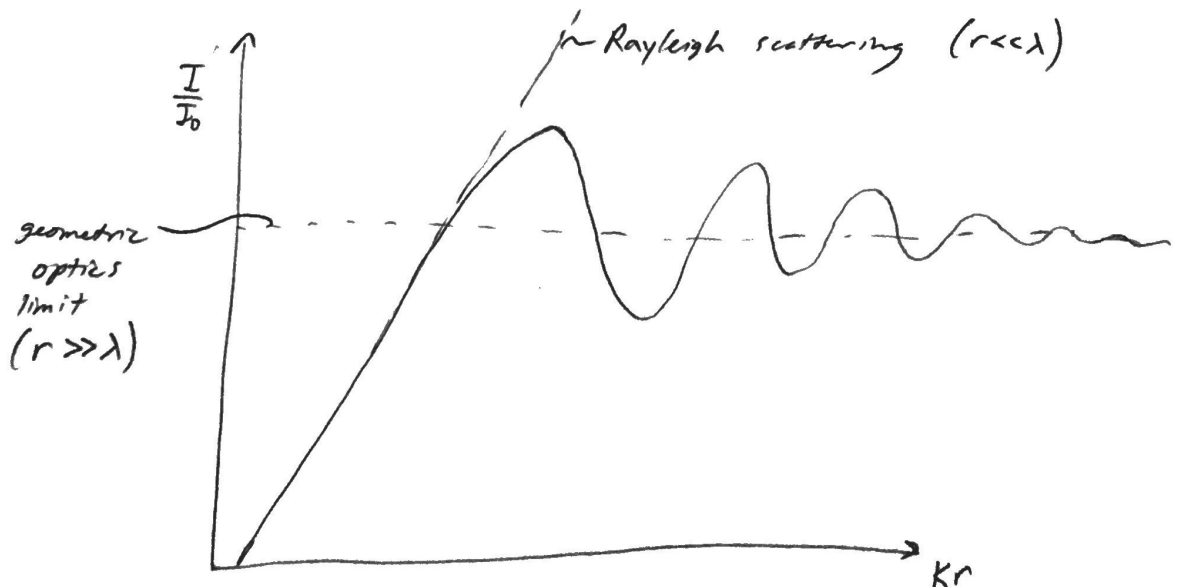
$$I = I_0 \left(\frac{\omega}{c} \right)^4 \frac{r^6}{R^2} \frac{1 + \cos^2 \phi}{2} \left(\frac{n^2 - 1}{n^2 + 2} \right)^2 \left(1 + C_1 K^2 r^2 + C_2 K^4 r^4 + \dots \right)$$

where

$$C_1 = \frac{6}{5} \left(\frac{n^2 - 1}{n + 2} \right)$$

$$C_2 = \frac{3}{175} \frac{n^6 + 41n^4 - 28n^2 + 284}{(n^2 + 2)^2} + \frac{1}{900} \left(\frac{n^2 + 2}{2n^2 + 2} \right)^2 [15 + (2n^2 + 3)^2]$$

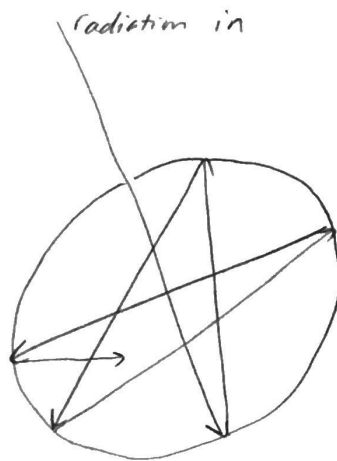
- It's clear from the above that when $r \ll \lambda$ ($Kr \ll 1$), the Mie scattering expression above reduces to the Rayleigh approximation.
- For $r \gg \lambda$, interference is present causing I to oscillate in frequency space. Conceptually



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Solar radiation at the top of the atmosphere

- Sun is well-represented as a black body (an idealized structure that absorbs all incident EM radiation regardless of frequency and incidence angle)
- Visualize black-body
Large cavity w/ opaque walls and a small hole



- hole is small $\Rightarrow$ escape unlikely
- at every reflection, radiation absorbed by wall
- eventually, equilibrium is reached between absorption, emission, and wall temperature

- To derive theory for cavity radiation, Planck assumed atoms in wall were tiny oscillators (just like we did in lecture 1) each w/ characteristic oscillation frequency
- Oscillators emit energy into and absorb energy from cavity

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- Planck made 2 assumptions about oscillators

1. Energy given by

$$E = \hbar \omega \left(n + \frac{1}{2} \right)$$

where n = principal quantum number
as we saw last lecture

2. Energy is quantized

$$\Delta E = \hbar \omega$$

- Determining emitted energy requires knowing total number of oscillators ω / freq ω for all possible states

- From Boltzmann statistics, number of oscillators ω energy

$$E > \tilde{E} \text{ is}$$

$$N = N_0 e^{\frac{-\tilde{U}}{k_B T}}$$

$k_B \equiv$ Boltzmann's const

$T \equiv$ absolute temperature

- Energy is quantized, thus $\tilde{U}$ takes on discrete values

$$\tilde{U} = 0, \hbar \omega, 2\hbar \omega, \dots$$

- Therefore, total number of oscillators ω / freq. ω

$$N_\omega = N_0 + N_0 e^{\frac{-\hbar \omega}{k_B T}} + N_0 e^{\frac{-2\hbar \omega}{k_B T}} + \dots$$

$$\approx \frac{N_0}{1 - e^{-\hbar \omega / k_B T}}$$

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- Total energy of oscillators comes from multiplying each term by appropriate energy level

$$E = 0 \cdot N_0 + \hbar\omega N_0 e^{\frac{-\hbar\omega}{k_B T}} + 2\hbar\omega N_0 e^{\frac{-2\hbar\omega}{k_B T}} + \dots$$

$$\approx \frac{\hbar\omega N_0 e^{\frac{-\hbar\omega}{k_B T}}}{(1 - e^{\frac{-\hbar\omega}{k_B T}})^2}$$

- Average energy per oscillator is

$$\frac{E}{N\omega} = \frac{\hbar\omega}{e^{\frac{\hbar\omega}{k_B T}} - 1}$$

- Now define monochromatic energy density u_ω (energy per volume per frequency)

$$u_\omega = A \frac{\hbar\omega}{e^{\frac{\hbar\omega}{k_B T}} - 1}$$

where $A = \frac{8\pi f^2}{c^3}$ with $f = \frac{\omega}{2\pi}$

$c \equiv$ speed of light

The value of A ensures that u_ω tends to the Rayleigh-Jeans law at low frequencies

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- Finally, define spectral radiance (power per unit solid angle per unit projected area $\Rightarrow$ units $\text{W sr}^{-1} \text{m}^{-2}$)

$$B_\omega = \frac{u_\omega c}{4\pi} = \frac{2hf^3}{c^2} \left[\exp\left\{\frac{hf}{k_B T}\right\} - 1 \right]^{-1}$$

which can be written in terms of wavelength as

$$B_\lambda = -B_\omega \frac{df}{d\lambda} = \frac{2hc^2}{\lambda^5} \left[\exp\left\{\frac{hc}{\lambda k_B T}\right\} - 1 \right]^{-1}$$

where

$h = 2\pi\hbar \equiv$ Planck's constant

- Take-homes

- Strong frequency (and wavelength) dependence

$\hookrightarrow$ peak spectral radiance at wavelength

$$\lambda_{\max} = \frac{b_W}{T} \quad \text{where } b_W = 2.898 \times 10^{-3} \text{ m} \cdot \text{K}$$

$\equiv$ Wein's displacement const

Known as Wein's displacement

$\hookrightarrow$ rapid drop off in radiance for $\lambda < \lambda_{\max}$ because it is hard for photons with energy $\gg k_B T$ (mean or characteristic energy) to be emitted

$\hookrightarrow$ long tail at $\lambda > \lambda_{\max}$ due to Boltzmann statistics

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- Strong T dependence $\rightarrow$ influences both total radiated energy and distribution

- For Sun ($T \approx 5800$ K at surface), highest radiance in violet-blue spectrum

$\hookrightarrow$ Recalling Rayleigh scattering intensity $\propto \omega^4$ we get blue skies (complemented by absorption by oxygen in atmosphere)

• Finally, in RS it is sometimes the case that we only measure the total radiant intensity

$$L(T) = \int_0^\infty B_\lambda(T) d\lambda$$

$$= \frac{2K_B^4 T^4}{h^3 c^2} \underbrace{\int_0^\infty \frac{z^3}{e^z - 1} dz}_{= \frac{\pi^4}{15}}$$

$$\text{where } z = \frac{hc}{\lambda K_B T}$$

$$\Rightarrow L(T) = b T^4 \quad \text{where } b = \frac{2\pi^4 K_B^4}{15 c^2 h^3} = \text{const}$$

- Since black body radiation is isotropic, intensity...

$$I = \pi L = \sigma T^4$$

$$\sigma = 5.67 \times 10^{-8} \left[\frac{W}{m^2 K^4} \right]$$

$\equiv$ Stefan - Boltzmann Const